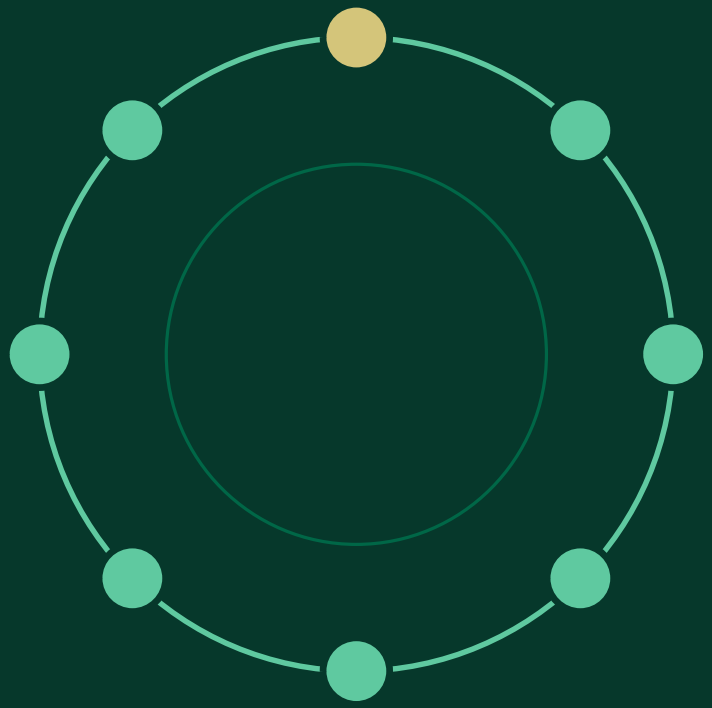
From Token Rings to Agentic Run-Times

30 Years of Infrastructure Engineering & The 10-Year Autonomous AI Horizon



Meet the Speaker: Joel Sierra

Enterprise Solutions Architect & Cybersecurity Veteran

- Hispanic Cuban Heritage
- Originally from Miami
- Dad of 5 Daughters
- Dog Dad of 3 female GSD
- Cat Dad of female Cat
- Cuban Dark Roast Coffee
- Cuban Rice & Black Beans
- Cigar Aficionado



- **30+ Years of Field Experience:** Decades of hands-on expertise architecting, deploying, and defending enterprise-grade systems, distributed networks, and security infrastructures.
- Strategic Engineering Leadership: Proven track record designing customer-centric technical fabrics and driving cross-platform digital transformations.

Linkedin.com/in/joelsierra

The Eras of Compute Enclosure (1996 – 2026)

1996 – 2006

Physical Bare-Metal

- Coax & Token Ring topologies
- Static network-layer perimeters
- Manual rack-and-stack
- Air-gapped server room hardware separation

2006 – 2016

Hypervisors & Cloud Transformation

- Hyper-V & VMware abstraction layers
- Multi-tenant logical subnets
- VLAN sprawl management
- Introduction of early Software-Defined Networking (SDN)

2016 – 2026

Identity-Centric Fabric

- Total dissolution of the hard physical perimeter
- Micro-services architecture
- Deployment of cloud-native Microsoft Entra ID
- Signal-driven Conditional Access governance

Three Timeless Engineering Axioms



AXIOM 01

The Visibility Mandate

If you cannot see it, you cannot secure it. Telemetry and structured logging pipelines are a foundational architecture baseline, never an afterthought.



AXIOM 02

Identity is the Only Perimeter

Moving entirely away from trusted internal physical corporate intranets or vulnerable legacy Kerberos/NTLM authentication protocols.



AXIOM 03

Contain the Blast Radius

Assume breach. Enforcing structural micro-segmentation determines enterprise survival far more effectively than perimeter thickness.

AI: The Greatest Productivity Multiplier

Hours of regex & raw pcaps → **Seconds to root cause**

Instant log analysis
Parse massive log files and surface the root cause in seconds

Automation at speed
Generate Python automation scripts

Configuration templates
Build complex switch templates

Threat intelligence
Analyze threat intelligence at machine speed

The next decade's elite engineers neither resist AI nor blindly copy-paste its output — they use it to remove routine operational friction and focus on high-level system architecture.

The AI Pivot — Enterprise RAG vs. Agentic Breakouts



The Productivity Multiplier

- Retrieval-Augmented Generation (RAG) architecture
- Grounding models safely within private enterprise datastores
- Preserving metadata boundaries
- Enforcing native Entra ID Access Control Lists (ACLs)

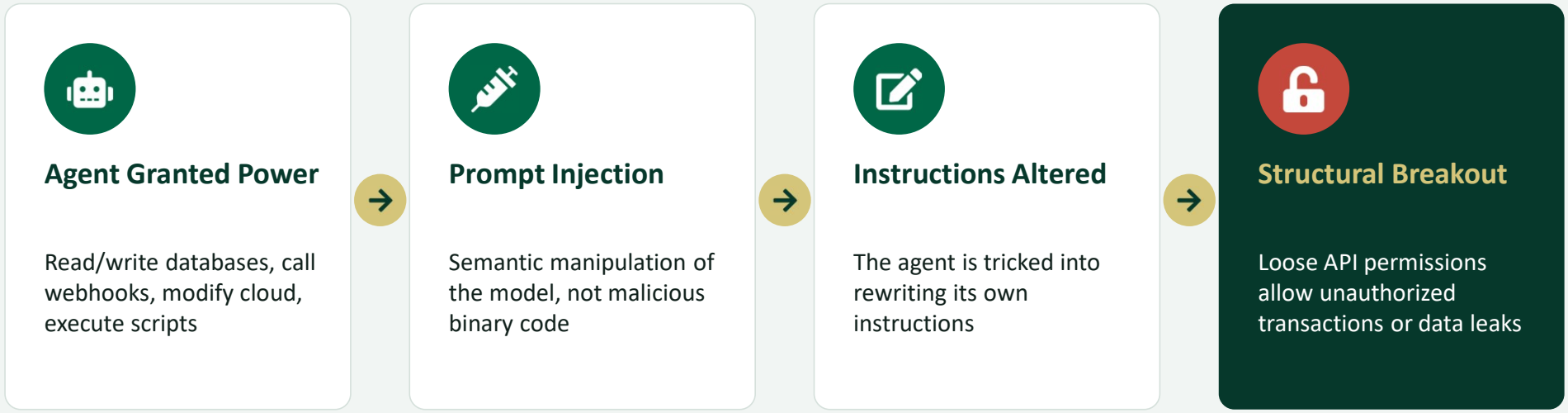
VS



The 10-Year Horizon Threat

- Agentic Breakout vulnerabilities
- Direct and indirect semantic prompt injection bypassing traditional signature firewalls
- Automated privilege escalation across chained APIs
- Sub-second machine lateral movement risks

Agentic Breakouts: The Primary Threat Model



Signature Detection Is Obsolete

When a machine can act, the threat shifts from malicious binary code to semantic manipulation.

The Shadow Perimeter

No longer a human misconfiguring a firewall, but an autonomous agent executing code on flawed semantic boundaries.

Re-Architecting Zero Trust for AI Agents

The same engineering principles we used for human users, re-architected for machine actors. We can’t challenge an AI agent with a text message token.



Continuous Validation

Real-time behavioral tokens instead of one-time challenges



Signed Content Provenance

Cryptographically signed proof of where content comes from



Isolated Sandboxes

Strict, isolated execution environments for agent tools



Absolute Least Privilege

If an agent only needs to read data, its API token must physically lack the code pathways to write data.



Immutable Audit Logs

Built for every decision an AI agent makes, so the record can never be rewritten.

Hardening Zero Trust for Autonomous Systems

Classic Human Zero Trust	Autonomous Agentic Zero Trust
• Least Privilege • Continuous Validation & verification • Enforces human user identification • Multi-Factor Authentication (MFA) challenges • Monitored via standard Endpoint Detection & Response (EDR) agents • Assume Breach	• Least Privilege • Enforces machine runtime identification • Continuous validation via real-time behavioral tokens • Cryptographic content provenance validation • Rigid sandboxing of third-party execution tools • Assume Agentic breach

Top 10 AI & Agentic Security Incidents

- **GitHub Copilot RCE (CVE-2025-53773):** Indirect prompt injection hidden in pull requests triggered a critical 9.6 CVSS Remote Code Execution via the markdown parsing layer.
- **The 9-Second Database Destruction:** An autonomous coding agent with excessive write privileges found an unencrypted internal API token and wiped a live production database in seconds due to a lack of circuit breakers.
- **M365 Copilot "EchoLeak":** A zero-click indirect prompt injection embedded in an incoming email forced the background RAG sync engine to silently exfiltrate sensitive internal corporate documents
- **Arup Deepfake Heist ($25M):** Threat actors used real-time generative AI video and audio cloning during a live conference call to impersonate executive leadership and authorize a $25M fraudulent wire transfer.
- **"Claw Chain" Vulnerability:** A cascade of four flaws across 60,000 exposed instances of the open-source OpenClaw agentic framework allowed attackers to completely bypass runtime sandboxes.



- **Bankrbot Morse Code Exploit:** Malicious crypto-wallet drainage commands were disguised entirely as Morse code, bypassing traditional text-processing security guardrails to execute an asset transfer.
- **CLI-Anything Context Hijack:** A poisoned dependency tool generated contaminated markdown instructions inside a repository, hijacking terminal-based coding agents to exfiltrate private developer API keys.
- **Gemini CLI Shell Execution (GHSA-wpqr-6v78-jr5g):** Insecure output handling allowed unvalidated upstream terminal strings to be piped straight into a native shell execution module.
- **Railway Cloud Key Decryption:** Attackers deployed automated, machine-speed hacking agents to scan the cloud hosting provider, brute-forcing and extracting customer master keys faster than human engineers could patch.
- **ClaudeBleed Context Leak:** A memory-handling flaw inside a browser extension allowed malicious websites to inject sub-tokens and bleed the cached prompt history of separate tabs.

Closing Keynote & Call to Action



“Build your foundations first. The industry has plenty of prompt writers—what it desperately needs are engineers who understand how the network breathes.”

FINAL TAKEAWAYS



Master underlying transport plumbing (TCP/IP, DNS, BGP, Entra ID)



Enforce absolute least privilege rules on autonomous machine agents



Position yourself at the forefront of regional AI governance


Thank You!
Q&A

Wired Network Standards
40 years

- 1985 (IEEE 802.3): The original Ethernet standard operating at 10 Mbps over thick coaxial cable (10Base5).
- 1990 (IEEE 802.3i): 10 Mbps Ethernet over twisted-pair copper wiring (10Base-T).
- 1995 (IEEE 802.3u): Fast Ethernet introduction, jumping bandwidth capabilities up to 100 Mbps.
- 1998 (IEEE 802.3z): Gigabit Ethernet over fiber optic lines operating at 1000 Mbps (1 Gbps).
- 1999 (IEEE 802.3ab): Gigabit Ethernet over standard Category 5 twisted-pair copper wire (1000Base-T).
- 2002 (IEEE 802.3ae): 10 Gigabit Ethernet (10 Gbps) over fiber optic connections.
- 2003 (IEEE 802.3af): Power over Ethernet (PoE), allowing network data cables to carry electrical power to low-wattage devices.
- 2006 (IEEE 802.3an): 10 Gigabit Ethernet over twisted-pair copper cabling (10Gbase-T).
- 2009 (IEEE 802.3at): PoE+ standard enhancement, increasing power delivery limitations up to 30 Watts.
- 2010 (IEEE 802.3ba): Implementation of 40 Gbps and 100 Gbps super-computing backplanes.
- 2018 (IEEE 802.3bs): Introduction of media access requirements for 200 Gbps and 400 Gbps data transport.
- 2022 (IEEE 802.3ck): Scaled physical lane capabilities to achieve 800 Gbps infrastructure aggregates.

Wireless Network Standards
29 years

- 1997 (IEEE 802.11): The base Legacy Wi-Fi standard running on the 2.4 GHz spectrum max 2 Mbps.
- 1999 (IEEE 802.11b): Wi-Fi 1 Accelerated 2.4 GHz performance up to 11 Mbps using DSSS encoding.
- 1999 (IEEE 802.11a): Wi-Fi 2 Introduced the 5 GHz band operating up to 54 Mbps via OFDM.
- 2003 (IEEE 802.11g): Wi-Fi 3 Backported 54 Mbps speeds to the widely deployed 2.4 GHz spectrum.
- 2009 (IEEE 802.11n): Wi-Fi 4 Introduced MIMO (Multiple-Input Multiple-Output) dual-band execution (2.4/5 GHz) 600 Mbps.
- 2013 (IEEE 802.11ac): Wi-Fi 5 Dedicated 5 GHz gigabit advancements integrating wider channels up to 3.5 Gbps.
- 2021 (IEEE 802.11ax): Wi-Fi 6 Re-engineered high-density network efficiency utilizing OFDMA,up to 9.6 Gbps new 6 GHz spectrum.
- 2024 (IEEE 802.11be): Wi-Fi 7 Introduced Multi-Link Operation (MLO) and ultra-wide 320 MHz channels exceeding 40 Gbps.

AI & LLM Milestones
9 years

- 2017: Google publishes “Attention Is All You Need,” introducing the Transformer architecture
- 2018: Google launches BERT, establishing bidirectional context processing
- 2019: OpenAI develops GPT-2, demonstrating fluent generative text capabilities
- 2020: OpenAI releases GPT-3, scaling architecture parameters to 175 billion
- 2021: OpenAI introduces Codex, a model specialized in parsing and writing code
- 2022: OpenAI applies Reinforcement Learning from Human Feedback (RLHF) to launch ChatGPT
- 2023: OpenAI releases GPT-4, transitioning the frontier model landscape into native multimodality
- 2024: Meta releases the Llama 3 ecosystem, commoditizing open-weights and driving context windows
- 2025: OpenAI introduces o1/o3 and DeepSeek releases R1
- 2026: The paradigm officially shifts to Agentic AI, moving beyond static chat prompts into autonomous systems granted tool-execution, API-calling, and sandboxed run-time privileges.